

Daphne Qin

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EDUCATION

Rice University - Houston, TX

Bachelor of Science in Computer Science, GPA 3.94/4.00

May 2027

SKILLS

Presentation: public speaking, conflict management

Programming: C, C++, Java, Python 3 (Flask, Matplotlib, scipy, scikit-learn, scikit-image), Go, JavaScript (Wix Velo, Discord.js), Typescript, NoSQL, SQLite3, Bash Script, Ubuntu Linux (WSL 2 shell), Git; Object-Oriented Programming

Languages: English (native proficiency), Mandarin Chinese (conversational)

PROFESSIONAL EXPERIENCE

Baylor College of Medicine - Houston, TX

Aiden Lab Researcher, Department of Molecular and Human Genetics

Jun 2024 - Present

- Discover 5 novel protein-DNA relationships to apply downstream to cancer research.
- Develop an efficient, large-scale pipeline heavily used by 10 professors and researchers to process millions of lines of previously untouched data using Python and Bash Script.
- Present discoveries to over 100 acclaimed researchers and laypeople at poster fairs and research conferences.

Bijanki Lab Research Assistant, Department of Neurosurgery

Jan 2024 - Present

- Develop a novel semi-supervised machine learning algorithm in Python for a 100% increase in efficiency when characterizing brain structures.
- Process electrical stimulation data from over 50 epilepsy patients undergoing brain activity monitoring to delineate brain connectivity variations in psychiatric disorders.

The Healthcare Navigator, American Medical Women's Association - Houston, TX

Associate Digital Engineer (Software Engineer)

Jan 2024 - Apr 2024

- Led the creation of a patient-tailored questionnaire by collaborating with a panel of 5 physicians.
- Reached college students nationwide to demystify the healthcare system and provide cost-effective healthcare provider suggestions for college students.
- Managed front- and backend using Wix Velo's JavaScript tech stack to neatly display recommended and affordable healthcare options based on user-selected preferences.

Memorial Sloan Kettering Cancer Center - New York, NY

Intern at Boire Lab, Human Oncology and Pathogenesis Program

Jul 2022 - Aug 2022

- Selected as one of 20 people, out of over 1000 applicants, to be chosen for this prestigious program.
- Conducted research on the effects of Id1 on leptomeningeal metastasis proliferation using cell culture and mouse model experiments.
- Conducted in-vitro cell viability assays under hypoxic/normoxic conditions to study the behavior of Id1 in different environments.
- Prepared mouse brain slides for imaging to detect Id1 movement within the leptomeninges.
- Analyzed data using Graphpad to quantify the effects of Id1 proliferation under different environments.

LEADERSHIP & ACTIVITIES

Rice Data Education and Exploration Program (DEEP) - Houston, TX

Mentor

Aug 2024 - Present

- Mentor a group of 11 students to choose and understand a public healthcare dataset using Python libraries, including Pandas, Matplotlib, and PyTorch.
- Support presentation of medical data insights to audiences of >100 faculty and healthcare professionals; team was awarded 2nd place for effective communication and discovery of correlation of commute time and smoking.

Data Analyst

Aug 2023 - Nov 2023

- Investigated correlations between BMI, HbA1C, and smoking in a public diabetes dataset with Python, Matplotlib, and scikit-learn.
- Presented findings to 75 industry officials; team was awarded 1st place at the program showcase for the discovery of the correlation of smoking with increased diabetes rates.

Rice Teaching Assistant - Houston, TX

Computational Thinking (Foundational Course for Computer Science Majors)

Aug 2024 - May 2025

- Coordinated a team of 50 TAs to organize an interactive, campus-wide treasure hunt to demonstrate the A-Star algorithm to students.
- Mentored over 300 students in concepts such as Markov chains, graph search algorithms, and predictive model fitting.

AWARDS & HONORS

President's Honor Roll (Fall 2023/2024), Certificate for Excellence in Technology Education (2023), Thomas Richard and Julia Hadley Franklin Scholarship (2023/2024)